

Empathy-GPT: Leveraging Large Language Models to Enhance Emotional Empathy and User Engagement in Embodied Conversational Agents

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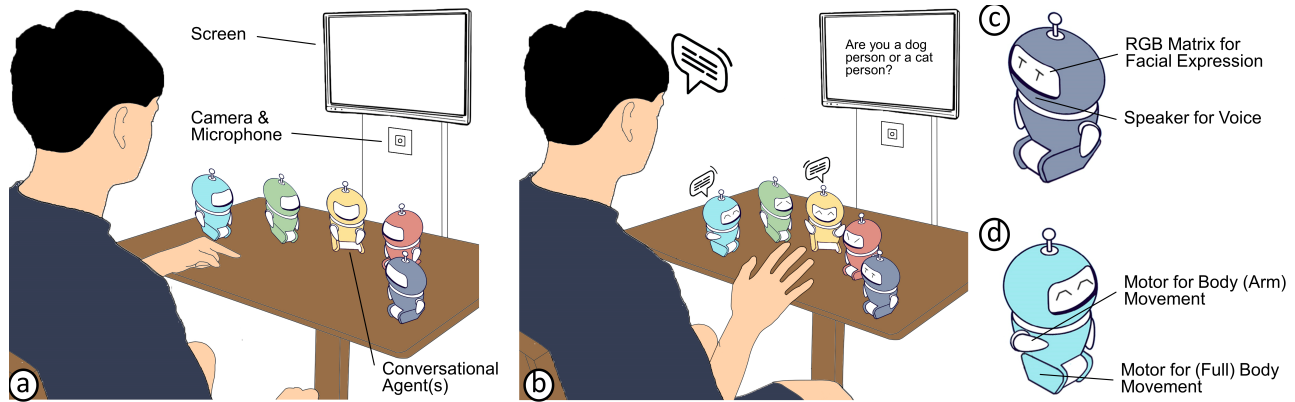


Figure 1: Figure 1 show components and interaction scenario of Empathy-GPT. (a) User's engage with embodied conversational agents. Components include a screen for displaying text, and a camera and microphone for capturing user input. (b) Interaction scenario where agents respond to the user through synchronized modalities (e.g., agents' voice, body movements, and facial expressions). (c) and (d) Detailed view of an agent equipped with an RGB matrix for facial expressions, a speaker for voice output, and motors for arm and full body movements, enabling expressive gestures.

ABSTRACT

Emotional empathy, the ability to understand and respond to others' emotions, is essential for effective communication. We propose Empathy-GPT, featuring embodied conversational agents with empathic capacity. To address the limitations of rule-based conversational agents, we leverage contextual understanding and adaptation capabilities of large language models (LLMs) to coordinate multiple modalities (e.g., agent's tone, body movements, and facial expressions). To enhance user engagement in human-agent communication, agents dynamically respond to users' voices and facial expressions, providing contextually empathic responses.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

LLM, Embodied Conversational Agent, Emotional Empathy

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1 INTRODUCTION

Emotional empathy, the ability to understand and respond to others' emotions, is essential for effective communication [14]. Embodied conversational agents with empathic capacity have long intrigued HCI researchers due to their ability to simulate human emotions and behaviors, maintain contextual conversations, and improve communication outcomes [9, 12]. However, developing empathic capabilities in these agents is complex, involving the coordination of multiple modalities (e.g., tone, body movements, and facial expressions) during conversations [1, 6].

Prior rule-based conversational agents, relying on pre-programmed commands, often struggle to coordinate these modalities, limiting the agents' expressiveness of emotions and behaviors and diminishing the conversational experience [2, 13]. Recent

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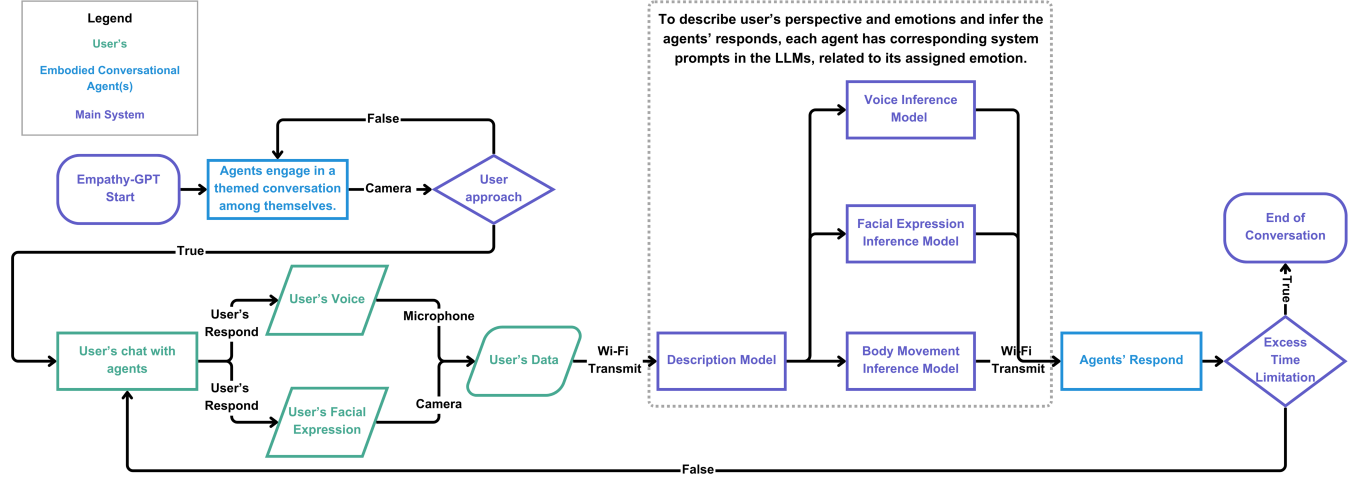


Figure 2: The flowchart of Empathy-GPT.

advancements in large language models (LLMs) offer new opportunities to address these challenges. LLMs provide superior contextual understanding, adaptation, and consistency in interactions, enabling better synchronization across all modalities [3, 5]. This makes them more effective in understanding and responding to human emotions [14].

We propose Empathy-GPT, an interactive system featuring embodied conversational agents with emotional empathy to enhance the quality of human-agent communication. By leveraging LLMs, these agents can understand users' emotions and provide empathic responses through more synchronized and consistent interactions across all modalities.

Inspired by prior research on emotion-driven interactions [4, 10, 11] and cinema [7, 8], we will design each agent to represent a specific emotion (e.g., Joy agent, Anger agent). During conversations, the agents will capture the user's voice and facial expressions and identify the emotions through LLMs. Each agent will then express corresponding empathic emotions and behaviors (e.g., changes in tone of voice, body movements, and facial expressions) based on the assigned emotion and conversational context.

For example, if the user's emotions align with the Joy agent, it might respond with a delighted tone, happy facial expressions, and enthusiastic body movements. Conversely, if the user's emotions oppose the Disgust agent, it might express a disappointed tone, turn its head, and move away. In this proposal, we empower embodied conversational agents to express and respond with empathic emotions and behaviors by utilizing LLMs to coordinate multiple modalities to enhance user engagement of human-agent communications.

2 EMPATHY-GPT AND ITS INTERACTION DESIGN

2.1 Empathy-GPT

The Empathy-GPT consist of two main modules: Embodied Conversational Agent(s) and Main System.

1. Embodied Conversational Agent(s): We will equip each agent with a microcontroller, microphone, speaker, RGB matrix, and motors. Each agent represents a specific emotion (i.e., Joy, Sad, Anger, Surprise, and Disgust) and has a corresponding color appearance. During the conversation, the microphone will collect the user's voice, and the speaker, RGB matrix, and motors will output the agent's voice, facial expressions, and body movements, respectively. The agents will respond based on the assigned emotion and conversational context from the main system.

2. Main System: It will receive data from each agent and collect the user's facial expressions using an additional Grove Vision AI Camera. Based on the corresponding system prompts for each agent's assigned emotion, the models (GPT-4) will infer how each agent should respond to the user (e.g., Anger agent might have an aggressive tone, furious facial expressions, and movement of approaching toward the user.)

2.2 Interaction Design

To encourage interaction with Empathy-GPT, the five LLM-driven agents will start a themed conversation among themselves. When users approach, the agents will invite them to join. Each agent will share their initial thoughts on the theme sequentially. Users can respond individually, interrupt, or listen to all thoughts before responding. During the conversation, Empathy-GPT will collect the user's voice and facial expressions, describe the user's perspective and emotions through the description LLM model, infer the corresponding responses (e.g., tone of voice, facial expressions, and body movements) through inference models, and signal specific agents. We hope that Empathy-GPT and its LLM-driven agents to create more engaging interactions through synchronized modalities.

3 POTENTIAL ADJUSTMENT FOR DEMO

After implementing Empathy-GPT, we will select the most feasible interaction design for a live demo. For example, we will consider user's cognitive load associated with the number of conversational agents and the complexity of their body movements.

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